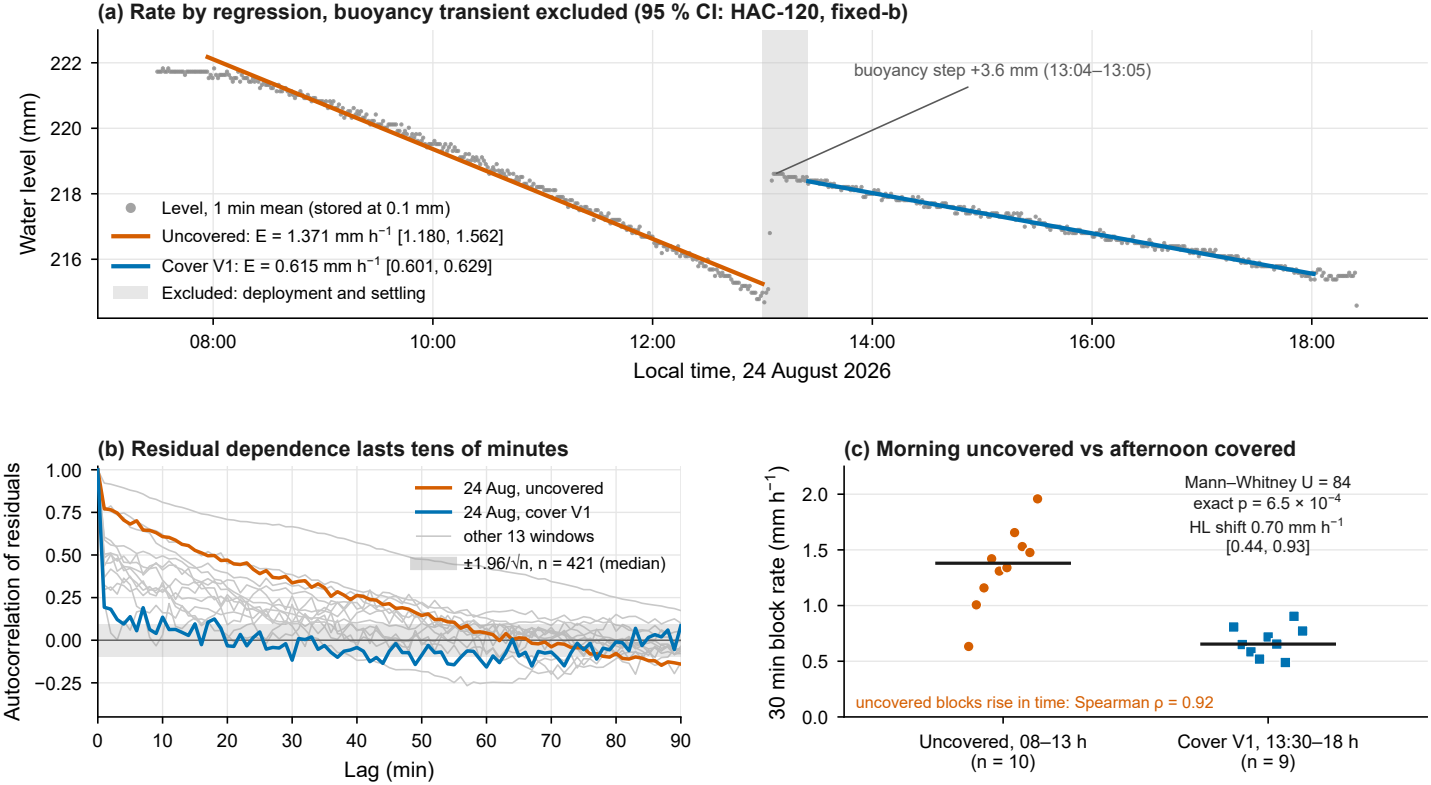

(a) Rate by regression, buoyancy transient excluded (95 % CI: HAC-120, fixed-b)
Water level (mm)
222
220
218
216
buoyancy step +3.6 mm (13:04–13:05)
Level, 1 min mean (stored at 0.1 mm)
Uncovered: E = 1.371 mm h⁻¹ [1.180, 1.562]
Cover V1: E = 0.615 mm h⁻¹ [0.601, 0.629]
Excluded: deployment and settling
08:00
10:00
12:00
14:00
16:00
18:00
Local time, 24 August 2026
(b) Residual dependence lasts tens of minutes
Autocorrelation of residuals
1.00
0.75
0.50
0.25
0.00
−0.25
24 Aug, uncovered
24 Aug, cover V1
other 13 windows
±1.96/√n, n = 421 (median)
0
10
20
30
40
50
60
70
80
90
Lag (min)
(c) Morning uncovered vs afternoon covered
30 min block rate (mm h⁻¹)
2.0
1.5
1.0
0.5
0.0
Mann–Whitney U = 84
exact p = 6.5 × 10⁻⁴
HL shift 0.70 mm h⁻¹
[0.44, 0.93]
uncovered blocks rise in time: Spearman ρ = 0.92
Uncovered, 08–13 h
(n = 10)
Cover V1, 13:30–18 h
(n = 9)